

Traian Florin Șerbănuță

List of Publications

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Publications

55 articles · h-index 22 · 2178 citations (Google Scholar, as of October 2026)

Reverse chronological within each category. Citation counts are from Google Scholar (October 2026); see also DBLP.

Journal Articles

1. Leuştean, Ioana, Natalia Moangă, and Traian Florin Șerbănuță. “Many-sorted hybrid modal languages.” *Journal of Logical and Algebraic Methods in Programming*, Vol. 120, 100644, 2021. [3]
2. Leuştean, Ioana, Natalia Moangă, and Traian Florin Șerbănuță. “A many-sorted polyadic modal logic.” *Fundamenta Informaticae*, Vol. 173, No. 2-3, pp. 191-215, 2020. [12]
3. Ștefănescu, Andrei, Ștefan Ciobâcă, Radu Mereuta, Brandon M Moore, Traian Florin Șerbănuță, and Grigore Roșu. “All-Path Reachability Logic.” *Logical Methods in Computer Science*, Vol. 15, Issue 2, 2019. [117]
4. Arusoaie, Andrei, Ștefan Ciobâcă, Dorel Lucanu, Grigore Roșu, Vlad Rusu, and Traian-Florin Șerbănuță. “Program Logics and Their Applications.” *Revue Roumaine de Mathématiques Pures et Appliquées*, Vol. 62, No. 1, pp. 137-154, 2017.
5. Rusu, Vlad, Dorel Lucanu, Traian-Florin Șerbănuță, Andrei Arusoaie, Andrei Ștefănescu, and Grigore Roșu. “Language Definitions as Rewrite Theories.” *Journal of Logical and Algebraic Methods in Programming*, Vol. 85, No. 1, pp. 98-120, 2016. [34]
6. Frei, Regina, Traian Florin Șerbănuță, and Giovanna Di Marzo Serugendo. “Self-organising assembly systems formally specified in Maude.” *Journal of Ambient Intelligence and Humanized Computing*, Vol. 5, No. 4, pp. 491-510, 2014. [12]
7. Frei, Regina, Giovanna Di Marzo Serugendo, and Traian Florin Șerbănuță. “Ambient intelligence in self-organising assembly systems using the chemical reaction model.” *Journal of Ambient Intelligence and Humanized Computing*, Vol. 1, No. 3, pp. 163-184, 2010. [26]
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9. Chira, Camelia, Traian Florin Șerbănuță, and Gheorghe Ștefănescu. “P systems with control nuclei: The concept.” *Journal of Logic and Algebraic Programming*, Vol. 79, No. 6, pp. 326-333, 2010. [9]
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11. Șerbănuță, Traian Florin, Grigore Roșu, and José Meseguer. “A Rewriting Logic Approach to Operational Semantics.” *Information and Computation*, Vol. 207, No. 2, pp. 305-340, 2009. [129]
12. Șerbănuță, Virgil Nicolae and Traian Florin Șerbănuță. “Injectivity of the Parikh matrix mappings revisited.” *Fundamenta Informaticae*, Vol. 73, No. 1-2, pp. 265-283, 2006. [63]
13. Șerbănuță, Traian Florin. “Extending Parikh matrices.” *Theoretical Computer Science*, Vol. 310, No. 1-3, pp. 233-246, 2004. [66]

Conference and Workshop Papers

1. Tușil, Jan, Traian Florin Șerbănuță, and Jan Obdržálek. “Cartesian Reachability Logic: A Language-parametric Logic for Verifying k-Safety Properties.” *Logic for Programming, Artificial Intelligence and Reasoning (LPAR 2023)*, pp. 405-456, 2023. [1]

2. Trufaş, Dafina, Ioan Teodorescu, Denisa Diaconescu, Traian Florin Şerbănuţă, and Vlad Zamfir. “Asynchronous Muddy Children Puzzle (work in progress).” *Working Formal Methods Symposium (FROM 2023)*, EPTCS Vol. 389, pp. 152-166, 2023.
3. Li, Elaine, Traian Florin Şerbănuţă, Denisa Diaconescu, Vlad Zamfir, and Grigore Roşu. “Formalizing Correct-by-Construction Casper in Coq.” *IEEE International Conference on Blockchain and Cryptocurrency (ICBC 2020)*, pp. 1-3, 2020. [19]
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5. Leuştean, Ioana, Natalia Moangă, and Traian Florin Şerbănuţă. “Operational Semantics and Program Verification Using Many-Sorted Hybrid Modal Logic.” *Automated Reasoning with Analytic Tableaux and Related Methods (TABLEAUX’19)*, 2019. [10]
6. Kasampalis, Theodoros, Dwight Guth, Brandon Moore, Traian Florin Şerbănuţă, Yi Zhang, Daniele Filaretti, Virgil Şerbănuţă, Ralph Johnson, and Grigore Roşu. “IELE: A Rigorously Designed Language and Tool Ecosystem for the Blockchain.” *Formal Methods (FM’19)*, pp. 593-610, 2019. [46]
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8. Daian, Philip, Dwight Guth, Chris Hathhorn, Yilong Li, Edgar Pek, Manasvi Saxena, Traian Florin Şerbănuţă, and Grigore Roşu. “Runtime Verification at Work: A Tutorial.” *Runtime Verification (RV’16)*, pp. 46-67, 2016. [12]
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11. Chiriţă, Claudia Elena and Traian Florin Şerbănuţă. “An Institutional Foundation for the K Semantic Framework.” *Recent Trends in Algebraic Development Techniques (WADT’14)*, LNCS Vol. 9463, pp. 9-29, 2015. [4]
12. Luo, Qingzhou, Yi Zhang, Choonghwan Lee, Dongyun Jin, Patrick O’Neil Meredith, Traian Florin Şerbănuţă, and Grigore Roşu. “RV-Monitor: Efficient Parametric Runtime Verification with Simultaneous Properties.” *Runtime Verification (RV’14)*, LNCS Vol. 8734, pp. 285-300, 2014. [108]
13. Ştefănescu, Andrei, Ştefan Ciobăcă, Radu Mereuta, Brandon M Moore, Traian Florin Şerbănuţă, and Grigore Roşu. “All-Path Reachability Logic.” *Rewriting and Typed Lambda Calculi (RTA-TLCA’14)*, LNCS Vol. 8560, pp. 425-440, 2014.
14. Arusoai, Andrei, Dorel Lucanu, Vlad Rusu, Traian-Florin Şerbănuţă, Andrei Ştefănescu, and Grigore Roşu. “Language Definitions as Rewrite Theories.” *Rewriting Logic and Its Applications (WRLA’14)*, LNCS, pp. 97-112, 2014.
15. Roşu, Grigore and Traian Florin Şerbănuţă. “K Overview and SIMPLE Case Study.” *Proceedings of K’11*, ENTCS Vol. 304, pp. 3-56, 2014. [64]
16. Şerbănuţă, Traian Florin, Andrei Arusoai, David Lazar, Chucky Ellison, Dorel Lucanu, and Grigore Roşu. “The K Primer (version 3.3).” *Proceedings of K’11*, ENTCS Vol. 304, pp. 57-80, 2014. [54]
17. Şerbănuţă, Traian Florin. “Rewriting Semantics and Analysis of Concurrency Features for a C-like Language.” *Proceedings of K’11*, ENTCS Vol. 304, pp. 167-182, 2014. [2]
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19. Şerbănuţă, Traian Florin. “Programming Language Semantics using K — true concurrency through term graph rewriting.” EPTCS Vol. 110, p. 2, 2013.
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21. Lazar, David, Andrei Arusoaie, Traian Florin Șerbănuță, Chucky Ellison, Radu Mereuta, Dorel Lucanu, and Grigore Roșu. "Executing Formal Semantics with the K Tool." *Formal Methods (FM'12)*, LNCS Vol. 7436, pp. 267-271, 2012. [21]
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27. Șerbănuță, Traian Florin, Gheorghe Ștefănescu, and Grigore Roșu. "Defining and Executing P Systems with Structured Data in K." *Membrane Computing (WMC'08)*, LNCS Vol. 5391, pp. 374-393, 2009. [28]
28. Chen, Feng, Traian Florin Șerbănuță, and Grigore Roșu. "jPredictor: a predictive runtime analysis tool for Java." *ICSE '08: Proceedings of the 30th International Conference on Software Engineering*, pp. 221-230, 2008. [155]
29. Ghoshal, Sudipto, Solaiappan Manimaran, Grigore Roșu, Traian Florin Șerbănuță, and Gheorghe Ștefănescu. "Monitoring IVHM Systems using a Monitor-Oriented Programming Framework." *Sixth NASA Langley Formal Methods Workshop (LFM 2008)*, 2008. [5]
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2. Șerbănuță, Traian Florin. *Programming Language Design and Analysis: A Rewriting Approach*. Editura Universității „Alexandru Ioan Cuza”, Iași, 2012. Revised edition of the PhD thesis.
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Preprints

1. Șerbănuță, Traian Florin, Jun Xu, Andrei Ștefănescu, and Cosmin Radoi. “Solving VeriContest with a Lean-Backed Rust Verifier.” arXiv:2610.03994, 2026.
2. Zamfir, Vlad, Mihai Calancea, Denisa Diaconescu, Wojciech Kołowski, Brandon Moore, Karl Palmskog, Traian Florin Șerbănuță, Michael Stay, Dafina Trușă, and Jan Tušil. “Validating Labelled State Transition and Message Production Systems: A Theory for Modelling Faulty Distributed Systems.” arXiv:2202.12662, 2022. [1]

Technical Reports

1. Arusoaie, Andrei and Traian Florin Șerbănuță. “Contextual Transformation in K Framework.” 2011. [3]
2. Șerbănuță, Traian Florin, Feng Chen, and Grigore Roșu. “Maximal Causal Models for Sequentially Consistent Systems.” University of Illinois, 2011.
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